Chapter 1

ACM for Real Yields

This chapter presents the model from Abrahams et al. (2016). The authors present a Gaussian affine term structure model (ATSM) for the joint pricing of the Treasury and TIPS yield curves that adjusts for the relative illiquidity of TIPS and generates estimates of the inflation risk premium.

1.1 The Model

1.1.1 Nominal Bonds

Assume the a $K \times 1$ vector of pricing factors evolves according to the autoregression

$$X_{t+1} = \mu_X + \Phi \left(X_t - \mu_X \right) + \nu_{t+1} \tag{1.1}$$

where ν_t are i.i.d. Gaussian with $E_t \left[\nu_{t+1} \right] = 0_{K \times 1}$ and $Var_t \left(\nu_{t+1} \right) = \Sigma$. Also assume that assets are priced by the SDF

$$M_{t+1} = \exp \left(-r_t - \frac{1}{2} \lambda_t' \lambda_t - \lambda_t' \Sigma^{-\frac{1}{2}} \nu_{t+1} \right) \tag{1.2}$$

where r_t is the nominal short rate and λ_t is the $K \times 1$ vector os prices of risk, which takes the essentially affine form

$$\lambda_t = \Sigma^{-\frac{1}{2}} \left(\lambda_0 + \lambda_1 X_t \right) \tag{1.3}$$

Under the risk neutral measure, the parameters that govern the dynamics of the pricing factors are

$$\tilde{\mu} = \left(I_K - \Phi \right) \mu_X - \lambda_0 \qquad \text{and} \qquad \tilde{\Phi} = \Phi - \lambda_1 \tag{1.4}$$

Denote $P_t^{(n)}$ as the price of the bond with maturity n at date t which, under the assumptions above, can be expressed as

$$\ln \left(P_t^{(n)} \right) = A_n + B_n' X_t \tag{1.5}$$

For $n = 0$, it is easy to see that $A_0 = 0$ and $B_0 = 0_{K \times 1}$. And for $n = 1$

$$\ln \left(P_t^{(1)} \right) = A_1 + B_1' X_t$$

$$-y_t^{(1)} = A_1 + B_1' X_t$$

$$r_t = -A_1 - B_1' X_t$$
$$r_t = \delta_0 + \delta_1' X_t \tag{1.6}$$

So we have that $A_1 = -\delta_0$ and $B_1 = -\delta_1$.

1.1.2 Affine Recursions for Nominal Bonds

We then impose a no-arbitrage restriction

$$P_t^{(n)} = E_t \left[M_{t+1} P_{t+1}^{(n-1)} \right]$$

Take logs on both sides

$$\ln\left(P_t^{(n)}\right) = E_t \left[\ln\left(M_{t+1}\right) + \ln\left(P_{t+1}^{(n-1)}\right) \right] + \frac{1}{2} Var_t \left[\ln\left(M_{t+1}\right) + \ln\left(P_{t+1}^{(n-1)}\right) \right] \tag{1.7}$$

First, lets look only at the term inside brackets

$$W_{t+1} = \ln\left(M_{t+1}\right) + \ln\left(P_{t+1}^{(n-1)}\right)$$

Substitute the log of equation (1.2) and equation (1.5)

$$W_{t+1} = -r_t - \frac{1}{2}\lambda_t'\lambda_t - \lambda_t'\Sigma^{-\frac{1}{2}}\nu_{t+1} + A_{n-1} + B_{n-1}' X_{t+1}$$

Substitute euqation (1.1)

$$W_{t+1} = -r_t - \frac{1}{2}\lambda_t'\lambda_t - \lambda_t'\Sigma^{-\frac{1}{2}}\nu_{t+1} + A_{n-1} + B_{n-1}' \left\{ \mu_X + \Phi\left(X_t - \mu_X\right) + \nu_{t+1} \right\}$$

Collect the common terms

$$W_{t+1} = -r_t - \frac{1}{2}\lambda_t'\lambda_t + A_{n-1} + B_{n-1}'\mu_X + B_{n-1}'\Phi\left(X_t - \mu_X\right) + \left(B_{n-1}' - \lambda_t'\Sigma^{-\frac{1}{2}}\right)\nu_{t+1}$$

The conditional mean of W_{t+1}

$$\begin{aligned} E_t\left[W_{t+1}\right] &= E_t\left[-r_t - \frac{1}{2}\lambda_t'\lambda_t + A_{n-1} + B_{n-1}'\mu_X + B_{n-1}'\Phi\left(X_t - \mu_X\right) + \left(B_{n-1}' - \lambda_t'\Sigma^{-\frac{1}{2}}\right)\nu_{t+1} \right] \\ &= -r_t - \frac{1}{2}\lambda_t'\lambda_t + A_{n-1} + B_{n-1}'\mu_X + B_{n-1}'\Phi\left(X_t - \mu_X\right) + \left(B_{n-1}' - \lambda_t'\Sigma^{-\frac{1}{2}}\right) E_t\left[\nu_{t+1}\right] \\ &= -r_t - \frac{1}{2}\lambda_t'\lambda_t + A_{n-1} + B_{n-1}'\mu_X + B_{n-1}'\Phi\left(X_t - \mu_X\right) \end{aligned} \tag{1.8}$$

The conditional variance of W_{t+1}

$$\begin{aligned}
Var_t\left[W_{t+1}\right] &= Var_t\left[-r_t - \frac{1}{2}\lambda_t'\lambda_t + A_{n-1} + B_{n-1}'\mu_X + B_{n-1}'\Phi\left(X_t - \mu_X\right) + \left(B_{n-1}' - \lambda_t'\Sigma^{-\frac{1}{2}}\right)\nu_{t+1}\right] \\
&= Var_t\left[\left(B_{n-1}' - \lambda_t'\Sigma^{-\frac{1}{2}}\right)\nu_{t+1}\right] \\
&= \left(B_{n-1}' - \lambda_t'\Sigma^{-\frac{1}{2}}\right) Var_t\left[\nu_{t+1}\right]\left(B_{n-1} - \Sigma^{-\frac{1}{2}}\lambda_t\right) \\
&= \left(B_{n-1}' - \lambda_t'\Sigma^{-\frac{1}{2}}\right)\Sigma\left(B_{n-1} - \Sigma^{-\frac{1}{2}}\lambda_t\right) \\
&= \left(B_{n-1}'\Sigma - \lambda_t'\Sigma^{-\frac{1}{2}}\Sigma\right)\left(B_{n-1} - \Sigma^{-\frac{1}{2}}\lambda_t\right) \\
&= B_{n-1}'\Sigma B_{n-1} - \lambda_t'\Sigma^{-\frac{1}{2}}\Sigma B_{n-1} - B_{n-1}'\Sigma\Sigma^{-\frac{1}{2}}\lambda_t + \lambda_t'\Sigma^{-\frac{1}{2}}\Sigma\Sigma^{-\frac{1}{2}}\lambda_t \\
&= B_{n-1}'\Sigma B_{n-1} - 2\lambda_t'\Sigma^{-\frac{1}{2}}\Sigma B_{n-1} + \lambda_t'\lambda_t \qquad (1.9)
\end{aligned}$$

Now we can substitute expressions (1.8) and (1.9) into equation (1.7)

$$\ln\left(P_t^{(n)}\right) = -r_t - \frac{1}{2}\lambda_t'\lambda_t + A_{n-1} + B_{n-1}'\mu_X + B_{n-1}'\Phi\left(X_t - \mu_X\right) + \frac{1}{2}\left(B_{n-1}'\Sigma B_{n-1} - 2\lambda_t'\Sigma^{-\frac{1}{2}}\Sigma B_{n-1} + \lambda_t'\lambda_t\right)$$

Canceling out terms

$$\ln\left(P_t^{(n)}\right) = -r_t + A_{n-1} + B_{n-1}'\mu_X + B_{n-1}'\Phi\left(X_t - \mu_X\right) + \frac{1}{2}B_{n-1}'\Sigma B_{n-1} - \lambda_t'\Sigma^{-\frac{1}{2}}\Sigma B_{n-1}$$

Substitute the definition of the price of risk from (1.3)

$$\ln\left(P_t^{(n)}\right) = -r_t + A_{n-1} + B_{n-1}'\mu_X + B_{n-1}'\Phi\left(X_t - \mu_X\right) + \frac{1}{2}B_{n-1}'\Sigma B_{n-1} - \left(\Sigma^{-\frac{1}{2}}\left(\lambda_0 + \lambda_1 X_t\right)\right)'\Sigma^{-\frac{1}{2}}\Sigma B_{n-1}$$

$$\ln\left(P_t^{(n)}\right) = -r_t + A_{n-1} + B_{n-1}'\mu_X + B_{n-1}'\Phi\left(X_t - \mu_X\right) + \frac{1}{2}B_{n-1}'\Sigma B_{n-1} - \left(\lambda_0 + \lambda_1 X_t\right)'\Sigma^{-\frac{1}{2}}\Sigma^{-\frac{1}{2}}\Sigma B_{n-1}$$

$$\ln\left(P_t^{(n)}\right) = -r_t + A_{n-1} + B_{n-1}'\mu_X + B_{n-1}'\Phi\left(X_t - \mu_X\right) + \frac{1}{2}B_{n-1}'\Sigma B_{n-1} - \left(\lambda_0 + \lambda_1 X_t\right)' B_{n-1}$$

$$\ln\left(P_t^{(n)}\right) = -r_t + A_{n-1} + B_{n-1}'\mu_X + B_{n-1}'\Phi\left(X_t - \mu_X\right) + \frac{1}{2}B_{n-1}'\Sigma B_{n-1} - \lambda_0' B_{n-1} - X_t'\lambda_1' B_{n-1}$$

Substitute the definition of r_t from (1.6) and transpose the last term, as it is a scalar.

$$\ln\left(P_t^{(n)}\right) = -\delta_0 - \delta_1' X_t + A_{n-1} + B_{n-1}'\mu_X + B_{n-1}'\Phi X_t - B_{n-1}'\Phi\mu_X + \frac{1}{2}B_{n-1}'\Sigma B_{n-1} - \lambda_0' B_{n-1} - B_{n-1}'\lambda_1 X_t$$

Collect the terms that involve X_t and the ones that do not

$$\ln\left(P_t^{(n)}\right) = \left(-\delta_0 + A_{n-1} + B_{n-1}'\mu_X - B_{n-1}'\Phi\mu_X + \frac{1}{2}B_{n-1}'\Sigma B_{n-1} - \lambda_0' B_{n-1}\right) + \left(-\delta_1' + B_{n-1}'\Phi - B_{n-1}'\lambda_1\right) X_t$$

$$\ln\left(P_t^{(n)}\right) = \left(A_{n-1} + B'_{n-1}\left(I-\Phi\right)\mu_X - B'_{n-1}\lambda_0 + \frac{1}{2}B'_{n-1}\Sigma B_{n-1} - \delta_0\right) + \left(B'_{n-1}\left(\Phi-\lambda_1\right) - \delta'_1\right)X_t$$

$$\ln\left(P_t^{(n)}\right) = \left(A_{n-1} + B'_{n-1}\left[\left(I-\Phi\right)\mu_X - \lambda_0\right] + \frac{1}{2}B'_{n-1}\Sigma B_{n-1} - \delta_0\right) + \left(B'_{n-1}\left(\Phi-\lambda_1\right) - \delta'_1\right)X_t$$

Recall the definitions of risk-neutral parameters from (1.4)

$$\ln\left(P_t^{(n)}\right) = \left(A_{n-1} + B'_{n-1}\tilde{\mu} + \frac{1}{2}B'_{n-1}\Sigma B_{n-1} - \delta_0\right) + \left(B'_{n-1}\tilde{\Phi} - \delta'_1\right)X_t$$

Notice that this last equation has the same form as equation (1.5). So we just found the affine recursions

$$A_n = A_{n-1} + B'_{n-1}\tilde{\mu} + \frac{1}{2}B'_{n-1}\Sigma B_{n-1} - \delta_0 \tag{1.10}$$

$$B'_n = B'_{n-1}\tilde{\Phi} - \delta'_1 \tag{1.11}$$

1.1.3 Excess Returns of Nominal Bonds

The definition of log-excess returns is

$$rx_{t+1}^{(n-1)} = \ln\left(P_{t+1}^{(n-1)}\right) - \ln\left(P_t^{(n)}\right) - r_t$$

Plug in equations (1.5) and (1.6)

$$rx_{t+1}^{(n-1)} = A_{n-1} + B'_{n-1}X_{t+1} - A_n - B'_nX_t - \delta_0 - \delta'_1X_t$$

$$rx_{t+1}^{(n-1)} = \left(A_{n-1} - A_n - \delta_0\right) - \left(B'_n + \delta'_1\right)X_t + B'_{n-1}X_{t+1}$$

Using the definition of the affine recursions in (1.10) and (1.11)

$$rx_{t+1}^{(n-1)} = -\left(B'_{n-1}\tilde{\mu} + \frac{1}{2}B'_{n-1}\Sigma B_{n-1}\right) - B'_{n-1}\tilde{\Phi}X_t + B'_{n-1}X_{t+1}$$

Now define $\alpha_{n-1} = -\left(B'_{n-1}\tilde{\mu} + \frac{1}{2}B'_{n-1}\Sigma B_{n-1}\right)$

$$rx_{t+1}^{(n-1)} = \alpha_{n-1} - B'_{n-1}\tilde{\Phi}X_t + B'_{n-1}X_{t+1} \tag{1.12}$$

As we will see later, this expression will be useful in the estimation.

1.1.4 Inflation-Indexed Bonds

We expand the ordinary Gaussian ATSM framework to allow for the pricing of inflation-indexed securities jointly with nominal securities so that both yield curves are affine in the state variables. Let Q_t be a price index at time t, and let $P_{R,t}^{(n)}$ denote the price at time t of an inflation-indexed bond with face value 1, paying out the quantity $\frac{Q_{t+n}}{Q_t}$ at time $t+n$. The price of such a bond satisfies

$$P_{R,t}^{(n)} = E_t^{\mathbb{Q}}\left[\exp\left(-\sum_{i=0}^{n-1} r_{t+i}\right)\frac{Q_{t+n}}{Q_t}\right]$$

where $E_t^{\mathbb{Q}}$ denotes the conditional expectation at time t under the pricing measure. Let one period log inflation be $\pi_t = \ln\left(\frac{Q_t}{Q_{t-1}}\right)$ such that

$$\frac{Q_{t+n}}{Q_t} = \exp\left(\sum_{i=1}^{n} \pi_{t+i}\right) \tag{1.13}$$

Just like nominal bonds, assume that log prices of inflation-indexed bonds are affine in the pricing factors:

$$P_{R,t}^{(n)} = A_{R,n} + B'_{R,n} X_t$$

Just like before it is straightforward that $A_{R,0} = 0$ and $B_{R,n} = 0_{K\times 1}$. Meaning that buying a real bond on the date of its maturity just gives you the price of 1 back. For the one period bond it has a face value of 1 at time t and will payout $\frac{Q_{t+1}}{Q_t}$ at time $t+1$. So its price today must satisfy

$$P_{R,t}^{(1)} = E_t^{\mathbb{Q}}\left[\exp\left(-r_t\right)\frac{Q_{t+1}}{Q_t}\right]$$

From the definition of the price index

$$P_{R,t}^{(1)} = E_t^{\mathbb{Q}}\left[\exp\left(-r_t + \pi_{t+1}\right)\right]$$

We already assumed that bond prices are affine in the state vector. For this assumption to still hold in the equation above, then we must make the assumption that

$$\pi_t = \pi_0 + \pi_1' X_t \tag{1.14}$$

In other words, equation (1.14) is a necessary consequence of the affine pricing assumption.

1.1.5 Affine Recursions for Real Bonds

We can impose the no-arbitrage condition

$$P_{R,t}^{(n)} = E_t^{\mathbb{Q}}\left[\exp\left(-r_t + \pi_{t+1}\right) P_{R,t+1}^{(n-1)}\right]$$

Taking logs and using the log-normal property

$$\ln\left(P_{R,t}^{(n)}\right) = E_t^{\mathbb{Q}}\left[-r_t + \pi_{t+1} + \ln\left(P_{R,t+1}^{(n-1)}\right)\right] + \frac{1}{2}Var_t^{\mathbb{Q}}\left[-r_t + \pi_{t+1} + \ln\left(P_{R,t+1}^{(n-1)}\right)\right]$$

$$\ln\left(P_{R,t}^{(n)}\right) = -r_t + E_t^{\mathbb{Q}}\left[\pi_{t+1} + \ln\left(P_{R,t+1}^{(n-1)}\right)\right] + \frac{1}{2}Var_t^{\mathbb{Q}}\left[\pi_{t+1} + \ln\left(P_{R,t+1}^{(n-1)}\right)\right] \tag{1.15}$$

Find the expected value term of equation (1.15)

$$\begin{aligned}
E_t^{\mathbb{Q}}\left[\pi_{t+1} + \ln\left(P_{R,t+1}^{(n-1)}\right)\right] &= E_t^{\mathbb{Q}}\left[\pi_0 + \pi_1' X_{t+1} + A_{R,n-1} + B'_{R,n-1} X_{t+1}\right] \\
&= \pi_0 + A_{R,n-1} + \left(\pi_1' + B'_{R,n-1}\right) E_t^{\mathbb{Q}}\left[X_{t+1}\right] \\
&= \pi_0 + A_{R,n-1} + \left(\pi_1 + B_{R,n-1}\right)' \left(\tilde{\mu} + \tilde{\Phi} X_t\right)
\end{aligned}$$

Find the variance term of equation (1.15)

$$\begin{aligned}
Var_t^{\mathbb{Q}}\left[\pi_{t+1}+\ln\left(P_{R,t+1}^{(n-1)}\right)\right] &= Var_t^{\mathbb{Q}}\left[\pi_0+\pi_1' X_{t+1}+A_{R,n-1}+B_{R,n-1}' X_{t+1}\right] \\
&= Var_t^{\mathbb{Q}}\left[\left(\pi_1+B_{R,n-1}\right)' X_{t+1}\right] \\
&= Var_t^{\mathbb{Q}}\left[\left(\pi_1+B_{R,n-1}\right)'\left(\tilde{\mu}+\tilde{\Phi}X_t+\nu_{t+1}\right)\right] \\
&= Var_t^{\mathbb{Q}}\left[\left(\pi_1+B_{R,n-1}\right)'\nu_{t+1}\right] \\
&= \left(\pi_1+B_{R,n-1}\right)' Var_t^{\mathbb{Q}}\left[\nu_{t+1}\right]\left(\pi_1+B_{R,n-1}\right) \\
&= \left(\pi_1+B_{R,n-1}\right)'\Sigma\left(\pi_1+B_{R,n-1}\right)
\end{aligned}$$

Substitute back into equation (1.15)

$$\ln\left(P_{R,t}^{(n)}\right) = -r_t+\pi_0+A_{R,n-1}+\left(\pi_1+B_{R,n-1}\right)'\left(\tilde{\mu}+\tilde{\Phi}X_t\right)+\frac{1}{2}\left(\pi_1+B_{R,n-1}\right)'\Sigma\left(\pi_1+B_{R,n-1}\right)$$

$$\ln\left(P_{R,t}^{(n)}\right) = -r_t+\pi_0+A_{R,n-1}+\pi_1'\tilde{\mu}+\pi_1'\tilde{\Phi}X_t+B_{R,n-1}'\tilde{\mu}+B_{R,n-1}'\tilde{\Phi}X_t+\frac{1}{2}\left(\pi_1+B_{R,n-1}\right)'\Sigma\left(\pi_1+B_{R,n-1}\right)$$

$$\ln\left(P_{R,t}^{(n)}\right) = -\delta_0-\delta_1' X_t+\pi_0+A_{R,n-1}+\pi_1'\tilde{\mu}+\pi_1'\tilde{\Phi}X_t+B_{R,n-1}'\tilde{\mu}+B_{R,n-1}'\tilde{\Phi}X_t+\frac{1}{2}\left(\pi_1+B_{R,n-1}\right)'\Sigma\left(\pi_1+B_{R,n-1}\right)$$

$$\ln\left(P_{R,t}^{(n)}\right) = -\delta_0-\delta_1' X_t+\pi_0+A_{R,n-1}+\pi_1'\tilde{\mu}+\pi_1'\tilde{\Phi}X_t+B_{R,n-1}'\tilde{\mu}+B_{R,n-1}'\tilde{\Phi}X_t+\frac{1}{2}\left(\pi_1+B_{R,n-1}\right)'\Sigma\left(\pi_1+B_{R,n-1}\right)$$

$$\ln\left(P_{R,t}^{(n)}\right) = \left(-\delta_0+\pi_0+A_{R,n-1}+\pi_1'\tilde{\mu}+B_{R,n-1}'\tilde{\mu}+\frac{1}{2}\left(\pi_1+B_{R,n-1}\right)'\Sigma\left(\pi_1+B_{R,n-1}\right)\right)+\left(-\delta_1'+\pi_1'\tilde{\Phi}+B_{R,n-1}'\tilde{\Phi}\right)X_t$$

$$\ln\left(P_{R,t}^{(n)}\right) = \left(\pi_0+A_{R,n-1}+\left(\pi_1+B_{R,n-1}\right)'\tilde{\mu}+\frac{1}{2}\left(\pi_1+B_{R,n-1}\right)'\Sigma\left(\pi_1+B_{R,n-1}\right)-\delta_0\right)+\left(\left(\pi_1+B_{R,n-1}\right)'\tilde{\Phi}-\delta_1'\right)X_t$$

This last equation is in the affine form we wanted it to be. So we get the affine recursions

$$A_{R,n} = \pi_0+A_{R,n-1}+\left(\pi_1+B_{R,n-1}\right)'\tilde{\mu}+\frac{1}{2}\left(\pi_1+B_{R,n-1}\right)'\Sigma\left(\pi_1+B_{R,n-1}\right)-\delta_0 \tag{1.16}$$

$$B_{R,n}' = \left(\pi_1+B_{R,n-1}\right)'\tilde{\Phi}-\delta_1' \tag{1.17}$$

1.1.6 Excess Returns of Real Bonds

The definition of log-excess returns of the real bonds is

$$rx_{R,t+1}^{(n-1)} = \ln\left(P_{R,t+1}^{(n-1)}\right)-\ln\left(P_{R,t}^{(n)}\right)-r_t$$

add inflation to both sides of the equation

$$rx_{R,t+1}^{(n-1)}+\pi_{t+1} = \ln\left(P_{R,t+1}^{(n-1)}\right)-\ln\left(P_{R,t}^{(n)}\right)-r_t+\pi_{t+1}$$

substitute the affine structure from (1.5), (1.14), (1.16) and (1.17)

$$rx_{R,t+1}^{(n-1)}+\pi_{t+1} = A_{R,n-1}+B_{R,n-1}'X_{t+1}-A_{R,n}-B_{R,n}'X_t-\delta_0-\delta_1' X_t+\pi_0+\pi_1' X_{t+1}$$

$$rx_{R,t+1}^{(n-1)} + \pi_{t+1} = (A_{R,n-1} - A_{R,n} + \pi_0 - \delta_0) + (B_{R,n-1} + \pi_1)' X_{t+1} - \left(B'_{R,n} + \delta'_1\right) X_t$$

Use equations (1.16) and (1.17) to simplify the intercept and the coeficient on X_t

$$rx_{R,t+1}^{(n-1)} + \pi_{t+1} = -\left[(\pi_1 + B_{R,n-1})' \tilde{\mu} + \frac{1}{2} (\pi_1 + B_{R,n-1})' \Sigma (\pi_1 + B_{R,n-1})\right] + (B_{R,n-1} + \pi_1)' X_{t+1} - (B_{R,n-1} + \pi_1)' \tilde{\Phi} X_t$$

Define $\alpha_{R,n-1} = -\left[(\pi_1 + B_{R,n-1})' \tilde{\mu} + \frac{1}{2} (\pi_1 + B_{R,n-1})' \Sigma (\pi_1 + B_{R,n-1})\right]$

$$rx_{R,t+1}^{(n-1)} + \pi_{t+1} = \alpha_{R,n-1} + (B_{R,n-1} + \pi_1)' X_{t+1} - (B_{R,n-1} + \pi_1)' \tilde{\Phi} X_t \tag{1.18}$$

1.1.7 Combining Nominal and Real Returns

We can stack the log-excess returns from equations (1.12) and (1.18) into a single vector R

$$R_{t+1} = \alpha - B\tilde{\Phi} X_t + B X_{t+1}$$

where

$$\alpha = -\left(B\tilde{\mu} + \frac{1}{2}\gamma\right)_{(N_N+N_R)\times 1}$$

$$B = \begin{pmatrix} B'_1 \\ \vdots \\ B'_{N_N} \\ (B_{R,1} + \pi_1)' \\ \vdots \\ (B_{R,N_R+\pi_1})' \end{pmatrix}_{(N_N+N_R)\times K}$$

$$\gamma = \begin{pmatrix} B'_1 \Sigma B_1 \\ \vdots \\ B'_{N_N} \Sigma B_{N_N} \\ (B_{R,1} + \pi_1)' \Sigma (B_{R,1} + \pi_1) \\ \vdots \\ (B_{R,N_R+\pi_1})' \Sigma (B_{R,N_R+\pi_1}) \end{pmatrix}_{(N_N+N_R)\times 1} \tag{1.19}$$

1.1.8 Expected Inflation and Inflation Premia

The affine model can be used to compute expected inflation under both the risk-neutral and the physical measure at any horizon. For a given forecast horizon n, model-implied average expected inflation under the pricing measure is given by the breakeven inflation rate, i.e., the difference between the yields on a nominal and an inflation-indexed bond with maturity n:

$$\begin{aligned} \pi_t^{(n)} &= y_t^{(n)} - y_{R,t}^{(n)} \\ &= -\frac{1}{n}\left[A_n + B'_n X_t - A_{R,n} - B'_{R,n} X_t\right] \end{aligned}$$

Model-implied expected average inflation under the physical measure is then simply obtained by replacing the parameters $\tilde{\mu}$ and $\tilde{\Phi}$ with their risk-adjusted counterparts, $(I_K - \Phi)\,\mu_X$ and Φ, in all of the affine recursions from equations (1.10), (1.11), (1.16) and (1.17). The difference between the two measures of inflation expectations constitutes the inflation risk premium, the compensation investors require for bearing inflation risk.

1.1.9 Liquidity Effects

It is straightforward to model the impact of liquidity on nominal and real yields in our framework. Let L_t be a factor capturing systematic changes in liquidity which we assume to be observed. We can simply expand the state space so that the vector of factors X_t includes this liquidity factor. The pricing coefficients remain subject to the recursive restrictions given in equations(1.10), (1.11), (1.16) and (1.17) above.

1.2 Estimation

1.2.1 Risk-Neutral VAR Parameters

If we have $N = N_N + N_R$ maturities, to estimate our parameters we stack the observed return data as

$$R = \alpha\iota_T' - B\tilde{\Phi}X_ + BX + E$$

where R is $N \times T$, X and $X_$ are $K \times T$ matrices of the stacked X_{t-1}'s and X_t's respectively, and ι_T is a $T \times 1$ vector of ones. The matrix E is the stacked return measurement errors (or return pricing errors) where the s-th column e_s satisfies $E_{s-1}[e_s] = 0$ and $E_{s-1}[e_s e_s'] = \Sigma_e$.

First run a seemingly unrelated regressions (SUR) of R on ι_T, $X_$ and X, which yields

$$\begin{pmatrix} \hat{\alpha}_{OLS} & \widehat{B\tilde{\Phi}}_{OLS} & \hat{B}_{OLS} \end{pmatrix}$$

Using the residuals $\hat{E}_{OLS}$ from this regression, we can estimate the pricing error covariance matrix as

$$\hat{\Sigma}_e = \frac{1}{T}\hat{E}_{OLS}\hat{E}_{OLS}'$$

So the estimator of $\tilde{\Phi}$ is

$$\hat{\tilde{\Phi}}_{GLS} = -\left(\hat{B}_{OLS}'\hat{\Sigma}_e^{-1}\hat{B}_{OLS}\right)^{-1}\hat{B}_{OLS}'\hat{\Sigma}_e^{-1}\widehat{B\tilde{\Phi}}_{OLS}$$

We the run the SUR on ι_T and $\hat{\tilde{\Phi}}_{GLS}X_ + X$, to obtain more efficient estimates $\hat{\alpha}_{GLS}$ and $\hat{B}_{GLS}$. Finally, we estimate $\tilde{\mu}$ as

$$\hat{\tilde{\mu}}_{GLS} = -\left(\hat{B}_{GLS}'\hat{\Sigma}_e^{-1}\hat{B}_{GLS}\right)^{-1}\hat{B}_{GLS}'\hat{\Sigma}_e^{-1}\left(\hat{\alpha}_{GLS} + \frac{1}{2}\hat{\gamma}_{GLS}\right)$$

where $\hat{\gamma}_{GLS}$ is constructed using $\hat{B}_{GLS}$ and $\hat{\Sigma}$ withb the same structure as expression (1.19).

The parameters $\tilde{\mu}$ and $\tilde{\Phi}$ are related to the market price of risk parameters λ_0 and λ_1 via the relationships

$$\tilde{\mu} = (I_K - \Phi)\,\mu_X - \lambda_0 \qquad \text{and} \qquad \tilde{\Phi} = \Phi - \lambda_1$$

Since pricing factors X are observed and follow the joint vector autoregression given by equation (1.1), the OLS estimator of μ_X is simply given by the sample mean of the factor X and the OLS estimator for Φ is obtained by

regressing the demeaned observations of X on their one period lags equation by equation. We stack the estimated inovations into the matrix $\hat{V}$ and construct an estimator of the state variable covariance matrix

$$\hat{\Sigma} = \frac{1}{T}\hat{V}\hat{V}'$$

Given the estimates $\hat{\mu}_X$ and $\hat{\Phi}$, we then obtain estimates for the market prices of risk parameters via

$$\hat{\lambda}_0 = \left(I_K - \hat{\Phi}\right)\hat{\mu}_X - \hat{\tilde{\mu}}_{GLS}$$
$$\hat{\lambda}_1 = \hat{\Phi} - \hat{\tilde{\Phi}}_{GLS}$$

Before we can compute the recursive bond pricing parameters for nominal bonds given in equations, we need to estimate the parameters δ_0 and δ_1 governing the nominal short rate. Since the nominal short rate is directly observed, this is simply achieved by performing an OLS regression of the short rate onto a constant and the vector of pricing factors.

As can be seen in the affine recursions of the real bonds, we also need to estimate the parameters π_0 and π_1 which govern the evolution of inflation as a function of the state variables. We now show that inflation-indexed bond yields are linear and quadratic functions of these two parameters, respectively. Therefore the information in the time series and cross-section of inflation-indexed bonds can be used to estimate them.

From equation (1.17) we have that

$$B'_{R,n} = \left(\pi_1 + B_{R,n-1}\right)'\tilde{\Phi} - \delta'_1$$

Starting from $n = 1$

$$B'_{R,1} = \pi'_1\tilde{\Phi} - \delta'_1$$
$$\left(B_{R,1} + \pi_1\right)' = \pi'_1\tilde{\Phi} + \left(\pi_1 - \delta_1\right)'$$

Now for $n = 2$

$$B'_{R,2} = \left(\pi_1 + B_{R,1}\right)'\tilde{\Phi} - \delta'_1$$
$$\left(B_{R,2} + \pi_1\right)' = \left(\pi_1 - \delta_1\right)'\tilde{\Phi} + \pi'_1\tilde{\Phi}^2 + \left(\pi_1 - \delta_1\right)'$$
$$\left(B_{R,2} + \pi_1\right)' = \left(\pi_1 - \delta_1\right)'\left(I + \tilde{\Phi}\right) + \pi'_1\tilde{\Phi}^2$$

for $n = 3$

$$B'_{R,3} = \left(\pi_1 + B_{R,2}\right)'\tilde{\Phi} - \delta'_1$$
$$\left(B_{R,3} + \pi_1\right)' = \left(\pi_1 - \delta_1\right)'\left(I + \tilde{\Phi} + \tilde{\Phi}^2\right) + \pi'_1\tilde{\Phi}^3$$

From here we can see the pattern

$$\left(B_{R,n} + \pi_1\right)' = \pi'_1\tilde{\Phi}^n + \left(\pi_1 - \delta_1\right)'\sum_{l=0}^{n-1}\tilde{\Phi}^l$$

$$\left(B_{R,n} + \pi_1\right)' = \pi'_1\tilde{\Phi}^n + \pi'_1\sum_{l=0}^{n-1}\tilde{\Phi}^l - \delta'_1\sum_{l=0}^{n-1}\tilde{\Phi}^l$$

$$(B_{R,n}+\pi_1)' = \pi_1' \sum_{l=0}^{n} \tilde{\Phi}^l - \delta_1' \sum_{l=0}^{n-1} \tilde{\Phi}^l$$

define $\psi_n = \sum_{l=0}^{n-1} \tilde{\Phi}^l$

$$(B_{R,n}+\pi_1)' = \pi_1'\psi_{n+1} - \delta_1'\psi_n \tag{1.20}$$

From equation (1.16) we have

$$A_{R,n} - A_{R,n-1} = \pi_0 + (\pi_1 + B_{R,n-1})'\,\tilde{\mu} + \frac{1}{2}(\pi_1 + B_{R,n-1})'\,\Sigma\,(\pi_1 + B_{R,n-1}) - \delta_0$$

We start by building a definition

$$A_{R,n} = A_{R,n}$$

$$A_{R,n} = (A_{R,n} - A_{R,n-1}) + A_{R,n-1}$$

$$A_{R,n} = (A_{R,n} - A_{R,n-1}) + (A_{R,n-1} - A_{R,n-2}) + A_{R,n-2}$$

given the fact that $A_{R,0} = 0$, we can simply subtract this term at the end. This gives us

$$A_{R,n} = \sum_{i=1}^{n} (A_{R,i} - A_{R,i-1})$$

$$A_{R,n} = \sum_{i=1}^{n} \pi_0 + (\pi_1 + B_{R,i-1})'\,\tilde{\mu} + \frac{1}{2}(\pi_1 + B_{R,i-1})'\,\Sigma\,(\pi_1 + B_{R,i-1}) - \delta_0$$

$$A_{R,n} = n(\pi_0 - \delta_0) + \sum_{i=1}^{n} (\pi_1 + B_{R,i-1})'\,\tilde{\mu} + \frac{1}{2}(\pi_1 + B_{R,i-1})'\,\Sigma\,(\pi_1 + B_{R,i-1})$$

Using what we found in equation (1.20)

$$A_{R,n} = n(\pi_0 - \delta_0) + \sum_{i=1}^{n} (\pi_1'\psi_i - \delta_1'\psi_{i-1})\,\tilde{\mu} + \frac{1}{2}(\pi_1'\psi_i - \delta_1'\psi_{i-1})\,\Sigma\,(\pi_1'\psi_i - \delta_1'\psi_{i-1})'$$

$$A_{R,n} = n(\pi_0 - \delta_0) + \left(\sum_{i=1}^{n} \pi_1'\psi_i\tilde{\mu}\right) - \left(\sum_{i=1}^{n} \delta_1'\psi_{i-1}\tilde{\mu}\right) + \left(\sum_{i=1}^{n} \frac{1}{2}(\pi_1'\psi_i - \delta_1'\psi_{i-1})\,\Sigma\,(\pi_1'\psi_i - \delta_1'\psi_{i-1})'\right)$$

$$A_{R,n} = n(\pi_0 - \delta_0) + \left(\pi_1' \sum_{i=1}^{n} \psi_i\tilde{\mu}\right) - \left(\delta_1' \sum_{i=1}^{n} \psi_{i-1}\tilde{\mu}\right) + \left(\frac{1}{2}\sum_{i=1}^{n} (\pi_1'\psi_i - \delta_1'\psi_{i-1})\,\Sigma\,(\pi_1'\psi_i - \delta_1'\psi_{i-1})'\right)$$

$$A_{R,n} = n(\pi_0 - \delta_0) + \left(\pi_1' \sum_{i=1}^{n} \psi_i\tilde{\mu}\right) - \left(\delta_1' \sum_{i=1}^{n} \psi_{i-1}\tilde{\mu}\right) + \left(\frac{1}{2}\sum_{i=1}^{n} (\pi_1'\psi_i\Sigma - \delta_1'\psi_{i-1}\Sigma)\left(\psi_i'\pi_1 - \psi_{i-1}'\delta_1\right)\right)$$

$$A_{R,n} = n(\pi_0 - \delta_0) + \left(\pi_1' \sum_{i=1}^{n} \psi_i\tilde{\mu}\right) - \left(\delta_1' \sum_{i=1}^{n} \psi_{i-1}\tilde{\mu}\right) + \left(\frac{1}{2}\sum_{i=1}^{n} \left(\pi_1'\psi_i\Sigma\psi_i'\pi_1 - 2\delta_1'\psi_{i-1}\Sigma\psi_i'\pi_1 + \delta_1'\psi_{i-1}\Sigma\psi_{i-1}'\delta_1\right)\right)$$

$$A_{R,n} = n\pi_0 - n\delta_0 + \left(\pi_1' \sum_{i=1}^{n} \psi_i\tilde{\mu}\right) - \left(\delta_1' \sum_{i=1}^{n} \psi_{i-1}\tilde{\mu}\right) + \frac{1}{2}\left(\sum_{i=1}^{n} \pi_1'\psi_i\Sigma\psi_i'\pi_1\right) + \frac{1}{2}\left(\sum_{i=1}^{n} \delta_1'\psi_{i-1}\Sigma\psi_{i-1}'\delta_1\right) - \left(\sum_{i=1}^{n} \delta_1'\psi_{i-1}\Sigma\psi_i'\pi_1\right)$$

We now collect terms that do not depend on the coefficients π_0 and π_1, the ones that depend linearly, and the ones that depend quadratically

$$\begin{aligned}
A_{R,n} = & -n\delta_0 - \delta_1' \left(\sum_{i=1}^{n} \psi_{i-1} \right) \tilde{\mu} + \frac{1}{2} \left(\sum_{i=1}^{n} \delta_1' \psi_{i-1} \Sigma \psi_{i-1}' \delta_1 \right) \\
& + \left(\pi_1' \sum_{i=1}^{n} \psi_i \tilde{\mu} \right) - \left(\sum_{i=1}^{n} \pi_1' \psi_i \Sigma \psi_{i-1}' \delta_1 \right) \\
& + \frac{1}{2} \left(\sum_{i=1}^{n} \pi_1' \psi_i \Sigma \psi_i' \pi_1 \right) \\
& + n\pi_0
\end{aligned}$$

$$\begin{aligned}
A_{R,n} = & \left[\frac{1}{2} \left(\sum_{i=1}^{n} \delta_1' \psi_{i-1} \Sigma \psi_{i-1}' \delta_1 \right) - \delta_1' \left(\sum_{i=1}^{n} \psi_{i-1} \right) \tilde{\mu} - n\delta_0 \right] \\
& + \pi_1' \left[\left(\sum_{i=1}^{n} \psi_i \tilde{\mu} \right) - \left(\sum_{i=1}^{n} \psi_i \Sigma \psi_{i-1}' \delta_1 \right) \right] \\
& + \pi_1' \left[\frac{1}{2} \sum_{i=1}^{n} \psi_i \Sigma \psi_i' \right] \pi_1 \\
& + n\pi_0
\end{aligned}$$

Thus, real yields may be written as

$$y_{R,t}^{(n)} = -\frac{1}{n} \left(A_{R,n} + B_{R,n}' X_t \right)$$

where

$$\begin{aligned}
-\frac{1}{n} A_{R,n} = & \left[\delta_1' \left(-\frac{1}{2n} \sum_{i=1}^{n} \psi_{i-1} \Sigma \psi_{i-1}' \right) \delta_1 + \delta_1' \left(\frac{1}{n} \sum_{i=1}^{n} \psi_{i-1} \right) \tilde{\mu} + \delta_0 \right] \\
& + \pi_1' \left[- \left(\frac{1}{n} \sum_{i=1}^{n} \psi_i \tilde{\mu} \right) + \left(\frac{1}{n} \sum_{i=1}^{n} \psi_i \Sigma \psi_{i-1}' \right) \delta_1 \right] \\
& - \pi_1' \left[\frac{1}{2n} \sum_{i=1}^{n} \psi_i \Sigma \psi_i' \right] \pi_1 \\
& - \pi_0
\end{aligned}$$

and

$$-\frac{1}{n} B_{R,n}' = \delta_1' \left(\frac{1}{n} \psi_n \right) + \pi_1' \frac{1}{n} \left(I - \psi_{n+1} \right)$$

So we can write the real yield as

$$
\begin{aligned}
y_{R,t}^{(n)} = & \left[\delta_1' \left(-\frac{1}{2n}\sum_{i=1}^{n}\psi_{i-1}\Sigma\psi_{i-1}'\right)\delta_1 + \delta_1'\left(\frac{1}{n}\sum_{i=1}^{n}\psi_{i-1}\right)\tilde{\mu} + \delta_0\right] \\
& + \pi_1'\left[-\left(\frac{1}{n}\sum_{i=1}^{n}\psi_i\tilde{\mu}\right) + \left(\frac{1}{n}\sum_{i=1}^{n}\psi_i\Sigma\psi_{i-1}'\right)\delta_1\right] \\
& - \pi_1'\left[\frac{1}{2n}\sum_{i=1}^{n}\psi_i\Sigma\psi_i'\right]\pi_1 \\
& - \pi_0 \\
& + \delta_1'\left(\frac{1}{n}\psi_n\right) + \pi_1'\frac{1}{n}\left(I - \psi_{n+1}\right)
\end{aligned}
$$

$$
\begin{aligned}
y_{R,t}^{(n)} = & \left[\delta_1' \left(-\frac{1}{2n}\sum_{i=1}^{n}\psi_{i-1}\Sigma\psi_{i-1}'\right)\delta_1 + \delta_1'\left(\frac{1}{n}\sum_{i=1}^{n}\psi_{i-1}\right)\tilde{\mu} + \delta_1'\left(\frac{1}{n}\psi_n\right) + \delta_0\right] \\
& + \pi_1'\left[\left(\frac{1}{n}\sum_{i=1}^{n}\psi_i\Sigma\psi_{i-1}'\right)\delta_1 - \left(\frac{1}{n}\sum_{i=1}^{n}\psi_i\tilde{\mu}\right) + \frac{1}{n}\left(I - \psi_{n+1}\right)\right] \\
& - \pi_1'\left[\frac{1}{2n}\sum_{i=1}^{n}\psi_i\Sigma\psi_i'\right]\pi_1 \\
& - \pi_0
\end{aligned}
\tag{1.21}
$$

This expression shows us that we can write the real yield as a linear-quadratic function of the parameters π_0 and π_1. Denote the function in equation (1.21) as $g\left(\pi_0, \pi_1 | n_i, t\right)$, then we can solve the following problem to estimate these parameters

$$
\hat{\pi}_0, \hat{\pi}_1 = \underset{\pi_0,\pi_1}{\arg\min} \sum_{i=1}^{N_R}\sum_{t=1}^{T}\left(y_{R,t}^{(n_i)} - g\left(\pi_0, \pi_1 | n_i, t\right)\right)^2
$$

Bibliography

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